OBJECTIVES AND SYNOPSIS

CEREBROVASCULAR DISEASES

CENTRAL NERVOUS SYSTEM TRAUMA

LEARNING OBJECTIVES:

1. Recognize the three major categories of CNS vascular disease:
   - Thrombotic occlusion
   - Embolic occlusion
   - Vascular rupture
2. Know the relationship between the location of ischemic injury to the brain and the vascular territories (the anterior, middle, and posterior cerebral arteries, and the vertebrobasilar system)
3. Know the definition of a transient ischemic attack (TIA)
4. Know the definition and etiology of a watershed (borderzone) infarct
5. Know the definition and predisposing factors of a lacunar infarct.
6. Know the relative roles of thrombosis and embolism in causing CNS infarcts
7. Describe sources of emboli to the CNS vasculature
8. Describe the process of organization of an infarct in the CNS
9. Explain the basis of secondary hemorrhage in a “pale” infarct; compare and contrast with primary intraparenchymal hemorrhage
10. Describe the pathophysiology of subcortical vascular dementia (Binswanger disease)
11. List common risk factors for and major sites of primary intraparenchymal hemorrhage:
    - Hypertension and Charcot-Bouchard microaneurysms:
      - Basal ganglia, thalamus, pons and cerebellum
    - Cerebral amyloid angiopathy:
      - Lobar hemorrhages in the cerebral cortex
12. Describe epidural, subdural, and subarachnoid hemorrhage and the most common clinical settings where they occur:
13. Describe the pathophysiology of saccular vs. atherosclerotic vs. mycotic vs. dissecting vs. traumatic aneurysms
14. Describe the pathophysiology of the most common vascular malformations
15. Describe cerebral contusion and coup-contre coup injury
16. Describe diffuse axonal injury (DAI) and the clinical situations where it is most commonly encountered
17. Recognize the CNS herniation syndromes:
    a. Subfalcine (cingulate)

b. Transtentorial (uncal herniation is a subtype))
c. Tonsillar

18. Explain the pathophysiologic basis of the two major types of neonatal brain injury, hemorrhages and infarcts, and the involvement of the germinal matrix
19. Describe the pathophysiology of "the shaken baby" syndrome

READING ASSIGNMENT: Robbins Basic Pathology, 11th Ed.; Chapter 21: Edema, Hypoxia, Ischemia, Infarction; Central Nervous System Trauma.

CEREBROVASCULAR DISEASES

CNS VASCULAR LESIONS

TIA
Anemic stroke
Hemorrhagic stroke
Aneurysms
Vascular malformations

TRANSIENT ISCHEMIC ATTACK (TIA)

- Tissue-based definition
- A consensus statements from the American Heart Association and American Stroke Association (AHA/ASA): transient ischemic attack (TIA) is defined as a transient episode of neurologic dysfunction caused by focal brain, spinal cord, or retinal ischemia, **without** acute infarction
- Symptoms depend on the area of the brain involved.
- Most frequent symptoms:
  - Temporary loss of vision (amaurosis fugax),
  - Difficulty speaking (aphasia)
  - Numbness or tingling (paresthesia), usually on one side of the body
  - Weakness on one side of the body (hemiparesis)
- Less common symptoms:
  - Impairment of consciousness
  - Dizziness, lack of coordination or poor balance
- Prelude to a stroke; one third of patients with TIAs develop clinically significant infarcts within 5 years.

INFARCT = anemic stroke

Etiology and pathogenesis:

- Caused by local interruption of blood flow.

- Most common form of cerebrovascular disease; anemic strokes constitute 70-80% of cerebrovascular accidents.
- Most common in patients >60 years of age.
- More common in males than females.
- Most common cause: atherosclerosis
- Most common predisposing factors: hypertension, diabetes mellitus, smoking.
- Most severe atherosclerotic lesions: internal carotid arteries, proximal middle cerebral arteries, basilar artery.
- Thrombosis near the carotid bifurcation or in the basilar artery common.
- Most common cause of an intracranial arterial occlusion is an embolus from the heart or from a more proximal arterial segment blocking a branch of the middle cerebral arteries.
- Vasculitis and trauma less common.
- Size and distribution depends on the speed of thrombosis and the presence of collateral circulation.
- Drop in systemic blood pressure may lead to an infarct even without complete vascular occlusion in severe atherosclerosis.

Morphology of anemic brain infarct:
First 4-12 hrs:
- No gross or microscopic change..

By 12-18 hrs:
- Pink neurons
- Neuronal changes similar to global ischemia, but accompanied by neutrophils

By 36-48 hrs:
- Swelling and softening of parenchyma
- Blurring of the demarcation between gray and white matter
- Hemorrhages

By 72 hrs:
- Phagocytosis of the dead cells results in sharper demarcation of the infarct; peak macrophages at 7-14 days.

By 1 month:
- Extensive phagocytosis results in further softening and **liquefaction**
- Development of irregular cavities.

By 6 months:
- Complete liquefaction
- A thin rim of subpial parenchyma supplied by leptomeningeal arteries is preserved over a cortical infarct (helps to differentiate an old infarct from an old contusion).

Ischemic penumbra:
- Infarct has a central core of irreversible ischemic necrosis.

- Surrounding the core there is a zone of borderline ischemic tissue, the ischemic penumbra, which receives collateral circulation.
- Prompt thrombolytic therapy or clot extraction to restore blood flow may prevent irreversible damage in the penumbra.
- The window for salvaging the penumbra is 3-4 hours.

Lacunar infarcts:
- Small (<1.5 cm in diameter) infarcts in the deeper parts of the brain (basal ganglia, thalamus, white matter) and in the brain stem.
- Account for 20 percent of all strokes.
- Caused by occlusion of deep penetrating branches of major cerebral arteries.
- Common in arteriolosclerosis (small vessel disease) associated with **hypertension** and **diabetes.**
- Do not cause life-threatening cerebral edema, but a small lacunar infarct (e.g., one involving the internal capsule) can cause as severe a neurological deficit as a much larger hemispheric infarct.
- Multiple lacunar infarcts are associated with ischemic white matter degeneration and dementia (Binswanger disease)

Cerebral Autosomal-Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy (**CADASIL)**
- The most common form of **hereditary stroke disorder**
- Caused by mutations of the ***Notch 3*** **gene on chromosome 19.**
- Belongs to a family of disorders called the **leukodystrophies**.
- The most common clinical manifestations are **migraine headaches** and **transient ischemic attacks or strokes**, which usually occur between **40 and 50 years of age**.
- MRI is able to detect signs of the disease years prior to clinical manifestation of disease.

INTRACRANIAL HEMORRHAGE

Classification:
Based on the partition of the intracranial space into four anatomic compartments:
- Brain parenchyma
- Subarachnoid space
- Subdural space
- Epidural space

Etiology:
- Hypertension (causes vasculopathy leading to formation of Charcot-Bouchard microaneurysms)
- Berry aneurysm

- Vascular malformations
- Trauma
- Primary and metastatic neoplasms
- Bleeding associated with hematologic malignancies
- Coagulation disorders and vasculitis

Spontaneous intraparenchymal hemorrhage causes hemorrhagic stroke:

- **Strong association with hypertension**
- Produces lesions predominantly in three locations:
    - 65% in the basal ganglia and thalamus
    - 15% in the pons
    - 10% in the cerebellum
- Large hematomas act as space occupying lesions and have the potential to produce focal neurologic deficits and to cause lethal **trans tentorial herniation.**
- Minute lesions (**petechiae**) are less harmful than the obstruction of the associated small vessel.

ANEURYSMS

Berry (saccular) aneurysm:

- Originates in a structurally weak area at the branching point of a large cerebral artery in or close to the circle of Willis
- >90% occur within the carotid supply.
- In 20-30% of cases aneurysms are multiple.
- Rupture is the most common cause of symptomatic subarachnoid hemorrhage in adults.

Charcot-Bouchard microaneurysm:

- Associated with hypertension
- Typical location: basal ganglia and thalamus, less often in pons and cerebellum

Atherosclerotic aneurysm:

- Fusiform.
- Typical location: internal carotid, vertebral, or basilar artery.
- Thrombosis causes a posterior stroke.

Mycotic (septic) aneurysm:

- Most commonly associated with subacute bacterial endocarditis.
- May be multiple.
- Due to weakening of the walls of usually small arteries by infected emboli.
- May cause hemorrhages anywhere in the brain.

- Infection may spread into the leptomeninges (leptomeningitis), or parenchyma resulting in an abscess/ abscesses.

VASCULAR MALFORMATIONS

AV malformation:

- **Congenital, low-resistance AV shunts** that siphon blood from the adjacent parenchyma.
- Distribution: cerebral hemispheres within the distribution of middle cerebral artery.
- Clinical features include seizures, neurologic deficits, and chronic mass effects.
- Rupture may cause an intraparenchymal or subarachnoid hemorrhage, or both.

Other (usually asymptomatic) vascular malformations:

- Telangiectasia
- Venous angioma
- Cavernous angioma

CNS TRAUMA

Birth trauma:

- Intracranial hemorrhage is common in premature babies due to deformation of the soft skull during delivery and the fragility of the immature brain
- The germinal matrix of the immature brain is sensitive to oxygen deprivation and lack of oxygen may result in infarcts.

Concussion:

- Mild form of diffuse axonal injury (DAI; vide infra)
- Repeated concussions cause chronic traumatic encephalopathy (please review previous lecture on CNS injury patterns)

Contusion:

- Coup-contrecoup injury **r**esults from a blow on the opposite side of the head.
- Ventral side of the frontal and temporal lobes are common locations

Subarachnoid hemorrhage:

- Rupture of a berry aneurysm and head trauma most common causes
- Clinical signs include: headache (increasing intracranial pressure), nuchal rigidity (meningeal irritation), alteration in mental status
- Complication: acute hydrocephalus due obstruction of the flow of CSF

Subdural hematoma:

- Results from trauma that shears the veins or small arteries that traverse the space between the arachnoid and the dura.
- Most vulnerable: superior cortical veins that lead to the superior sagittal sinus.

Chronic subdural hematoma:

- May occur after a trivial head trauma in the young and the old.
- Slowly accumulating hematoma becomes "localized" = surrounded by a membrane of fibrovascular tissue.
- Degenerating blood clot increases in size by drawing fluid from the CSF and by further hemorrhages from the fragile vessels of the organizing granulation tissue.
- Clinical symptoms and signs appear as a result of increased intracranial pressure.

Epidural hemorrhage:

- No space normally between the skull and dura
- Skull fracture shears the middle meningeal artery resulting in bleeding and dissection of the dura from the inner table of the skull
- The expanding mass increases intracranial pressure and may produce **herniation**
- Prompt evacuation is necessary to save the patient's life.

DIFFUSE AXONAL INJURY (DAI)

Etiology:

- Occurs most frequently in motor vehicle accidents
- Caused by blows to unsupported head

Mechanism:

- Cerebrum moves back and forth, pivoting around the upper brainstem, which together with cerebellum is held firmly fixed by the tentorium
- Falx cerebri prevents side-to-side motion

Clinical features:

- Patients with severe DAI become unconscious immediately after the injury and either remain comatose or go into a persistent vegetative state
- 50% of patients who develop coma shortly after trauma are thought to have white matter damage and DAI

Pathophysiology:

- Axons are stretched but not severed

- Sudden deformation causes compaction of neurofilaments, loss of microtubules and an arrest of the fast axoplasmic flow
- Mitochondria and other organelles accumulate proximal to the lesion and cause axonal swelling
- Axons may rupture several hours later
- Initial event is influx of calcium through the stretched axolemma
- The swellings are located at nodes of Ranvier where the axolemma is more liable to deform because there is no myelin
- Brain damage is most severe along midline structures (corpus callosum, brainstem) where the shear forces are greatest, and at the cortex-white matter junction because of the change in the consistency of brain tissue

CNS HERNIATION SYNDROMES

Definition:

- Displacement of brain tissue past rigid dural folds (the falx and tentorium) or through openings in the skull.

Pathophysiology:

- A focal (tumor, abscess, hemorrhage) or diffuse (generalized brain edema, diffuse infection) space occupying process inside the scull displaces the CSF and compresses the vasculature and causes intracranial pressure to increase.
- Elevated intracranial pressure reduces perfusion of the brain and further exacerbates cerebral edema.
- When the increase is beyond the limit permitted by compression of veins and displacement of CSF, tissue herniates between compartments across the pressure gradient.
- If the expansion is sufficiently severe, herniation may occur in multiple anatomic locations

Variants:

Subfalcine (cingulate) herniation:

- Unilateral or asymmetric expansion of a cerebral hemisphere displaces the cingulate gyrus under the falx cerebri.
- May cause compression of the anterior cerebral artery and its branches.

Trans tentorial (uncinate, mesial temporal) herniation:

- Medial aspect of the temporal lobe is compressed against the free margin of the tentorium.
- The third cranial nerve is compressed, resulting in dilation of the pupil and impairment of ocular movements on the ipsilateral side.

- The posterior cerebral artery may be compressed, resulting in ischemic injury to the PCA territory, which includes the primary visual cortex.
- A large herniation may compress the contralateral cerebral peduncle (Kernohan's notch) resulting in ipsilateral hemiparesis.
- Progression of trans tentorial herniation causes distortion and tearing of penetrating veins and arteries supplying the upper brainstem resulting in flame-shaped hemorrhages in the midline and paramedian regions of the midbrain and pons (Duret hemorrhages).

Tonsillar herniation:
- Displacement of the cerebellar tonsils through the foramen magnum.
- Life-threatening because the resulting brainstem compression damages vital respiratory and cardiac centers in the medulla.

THE SHAKEN BABY SYNDROME

Prevalence:
- Child abuse is the second most common cause of death in children under one year of age, after the sudden infant death syndrome
- Estimated 50,000 cases of shaken baby syndrome in the United States each year
- Nearly 25% of infants with shaken baby syndrome die from the brain injuries sustained
- The victims range in age from just a few days to five years, with an average age of six to eight months
- Young men (early 20s) are the usual perpetrators; if the perpetrator is female, she is usually the caregiver

Pattern of injuries:
- Combination of subdural hematoma, subarachnoid hemorrhage, contusions, DAI, and retinal hemorrhage are often present
- Infants with these injuries are usually brought to the ER with a nonspecific clinical picture of hypotonia, listlessness, vomiting, irritability, and lethargy
- Diagnosis may be missed unless a skeletal survey and fundoscopic examination are done

Mechanism and pathogenesis:

- Shaking sets the head into a dangling motion and can cause whiplash spinal cord injury
- Movement of the brain inside the skull causes subdural and subarachnoid hemorrhage, contusions, white matter hemorrhages (gliding hemorrhages), tears of the corpus callosum, and DAI
- Injuries are a combination of shaking and blunt impact, thus, the term "shaking-impact syndrome" is more accurate
- Retinal hemorrhages and hemorrhages within the optic nerve sheaths are caused by shaking of the globe and tearing of delicate retinal vessels

Significance of retinal and optic nerve injury:

- Retinal and optic nerve injury are an important component of the shaken baby syndrome and correlate with the severity of brain damage
- Retinal hemorrhages are extremely uncommon in accidental brain injury
- Documentation of retinal hemorrhage by fundoscopic examination is important in the clinical and medico-legal evaluation of suspected child abuse cases

Prognosis:

- Usually poor; 20% of cases result in death within the first few days
- Survivors will most often be left with intellectual and developmental disabilities such as mental retardation or cerebral palsy
- Damage to the eyes can cause partial or total loss of vision
- Specialized care is usually required for the rest of the child’s life

KEY FACTS ABOUT CNS VASCULAR DISEASE AND TRAUMA:

- Cerebrovascular disease is the third most common cause of death in the US after heart disease and cancer.

- Arteriovenous malformations predispose to intracranial hemorrhage.

- Cerebral blood vessels are prone to atheroma, arteriolosclerosis and amyloid deposition.

- Berry aneurysms are the most common type of aneurysm of cerebral arteries.

- Stroke occurs in 2 per 1000 of the general population.

- The four types of ischemic stroke include: large vessel, small vessel, venous, and global.

- Regional cerebral infarction is caused by occlusion of named cerebral arteries.

- “Pale” or “anemic” infarct is caused by vascular occlusion.

- Lysis of a vascular occlusion and reperfusion (including treatment-related) may result in hemorrhage and a hemorrhagic “red” infarct.

- Lacunar infarcts are caused by arteriolosclerosis of small vessels and are common in hypertension and diabetes.

- Multiple lacunar infarcts can lead to subcortical vascular dementia (Binswanger disease)

- Cortical pseudolaminar necrosis is caused by generalized failure of perfusion (hypoxic/ischemic encephalopathy).

- Venous infarction is caused by venous sinus thrombosis.

- Intraparenchymal cerebral hemorrhage is most commonly caused by hypertensive vascular damage.

- Subarachnoid hemorrhage is most often caused by a ruptured berry aneurysm.

- Diffuse petechial hemorrhage in the brain is caused by damage to small vessels.

- Closed head injury (non-missile trauma) is caused by acceleration/deceleration forces to the head and results in rotation/shearing of the brain.

- Open head injury (missile trauma) is caused by penetration of the skull or brain by an external object (e.g. bullet).

- Cerebral contusion occurs when the brain moves within the cranial cavity causing parts of the brain to be crushed against the skull. Coup lesions occur at the site of impact, contre coup lesions on the opposite side.

- Diffuse axonal injury (DAI) is the result of shearing of axons due to acceleration/ deceleration/ torsion leading to severe damage to white matter tracts.

- Epidural hemorrhage/ hematoma are caused by tearing of vessels running between the skull and the dura. Middle meningeal artery is a common source, often associated with fracture of the temporal bone.

- Acute subdural hematoma is a result of a severe head injury (often skull fracture) and rupture of bridging veins and is usually associated with other types of brain injury.

- Chronic subdural hematoma occurs as a result of a minimal trauma and is seen in the very young and the elderly.

THE END